Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

Section A **3*10=30**

1)	Define a graph. What are the main components of a graph? CO1, RBT Level: 2	**3**	(CO1, L2)
2)	State the Handshaking Theorem and explain its significance. CO1, RBT Level: 3	**3**	(CO1, L3)
3)	What is a spanning tree? Explain with an example. CO1, RBT Level: 2	**3**	(CO1, L2)
4)	Define a Hamiltonian cycle. Give an example of a graph that has a Hamiltonian cycle and another one that does not. CO1, RBT Level: 3	**3**	(CO1, L3)
5)	Describe the difference between a complete graph and a regular graph. CO1, RBT Level: 2	**3**	(CO1, L2)
6)	Define a bipartite graph and provide an example. CO1, RBT Level: 3	**3**	(CO1, L3)
7)	What is a Eulerian path? Give an example of a graph with an Eulerian path and another one without. CO1, RBT Level: 4	**3**	(CO1, L4)
8)	Explain the concept of graph isomorphism. Give an example to illustrate your explanation. CO2, RBT Level: 3	**3**	(CO2, L3)
9)	Prove that every tree with n vertices has exactly n-1 edges. CO3, RBT Level: 4	**3**	(CO3, L4)

Section B **7*3=21**

11) a)	Prove that in any graph, the sum of the degrees of all vertices is twice the number of edges. CO3, RBT Level: 4	**7**	(CO3, L4)
12) a)	Prove that a graph is bipartite if and only if it contains no odd cycles. CO2, RBT Level: 5	**7**	(CO2, L5)

Section C **7*4=28**

AI-assisted draft - teacher review and approval required before use

Section D 7*3=21

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create